Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 7

Question 1

Given a set $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ of points in the plane, what is best running time for finding the $\sqrt{n}$ points closest to the origin (0,0) (using normal Euclidean distance)?

- $\Theta(\sqrt{n})$
- $\Theta(n)$
- $\Theta(n \log n)$
- $\Theta(n\sqrt{n})$

Question 1

Answer: B $\boldsymbol{\Theta}(n)$

Algorithm

- Compute distances to origin: $\boldsymbol{\Theta}(n)$
- Run Select to find $\sqrt{n}$-th element: $\boldsymbol{\Theta}(n)$
- Run Partition to get set of $\sqrt{n}$-th smallest elements: $\boldsymbol{\Theta}(n)$
- Total: $\boldsymbol{\Theta}(n)$

This is also the best possible as we need to at least look at every element: $\boldsymbol{\Omega}(n)$ time.

Question 2

It is possible to modify quicksort to run in **worst case time** of $\Theta(n \lg n)$ by changing the pivot selection method.

- True
- False

Question 2

Answer: True

- Linear time select to find median: $\Theta(n)$
- Partition: $\Theta(n)$
- Total time for divide and combine: $\Theta(n)$
- Total: $T(n) = 2T(n/2)+\Theta(n) = \Theta(n \log n)$

Question 3

Consider the following three algorithms to obain the k smallest numbers in sorted order:

I: Sort the numbers. Then list out the first k numbers.

II: Build a min-heap. Then run Extract-Min k times to get the smallest k elements.

III: Use order-statistic algorithm to find the kth smallest number. Partition around that number. Sort the k smallest numbers after partitioning.

Order the algorithms by slowest to fastest asymptotically in n and k, assuming comparison-based algorithms are used.

- I, II, III
- III, II, I
- II, I, III
- I, III, II
- II, III, I

Question 3

Answer: I, II, III

I. Sort then list out k numbers
   a) Sort: $\Theta(n \log n)$
   b) List out k numbers: $\Theta(k)$
   c) Total runtime: $\Theta(n \log n + k)$

II. Build min-heap. Extract-min k times.
   a) Build min-heap: $\Theta(n)$
   b) Extract min: $\Theta(k \log n)$
   c) Total runtime: $\Theta(n + k \log n)$

III. Run select to find k-th smallest number. Run partition. Sort the k smallest numbers.
   a) Select: $\Theta(n)$
   b) Partition: $\Theta(n)$
   c) Sort: $\Theta(k \log k)$
   d) Total runtime: $\Theta(n + k \log k)$

Question 4

Which algorithm should you select for finding the median in practice, if you are minimizing the expected runtime, and know that the input is selected by an adversary?

- Randomized quickselect
- Worst-case linear time Select

Question 4

Answer: A

- The adversary cannot affect the expected runtime of randomized select.
- Both algorithms have linear expected runtime.
- So, it comes down to the constant multiplier for the linear term. Randomized select has a much better constant and is almost always preferred in practice.